Supplemental Material

S1 Numerical simulations

In our numerical examples we simulate the Cartesian components $a \in \{x, y, z\}$ of the displacement field

$$\boldsymbol{u}(t) = \big(u^x(t), u^y(t), u^z(t)\big)^{\mathrm{T}}$$

of a Hookean solid that obeys the macroscopic overdamped elastodynamic model of the form

$$\frac{\bar{n}}{\sqrt{\bar{g}}}\delta(\xi-1)\bar{\gamma}\partial_t\boldsymbol{u} = -\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_{\mathrm{ela}} + \boldsymbol{f}_{\mathrm{flu}} + \boldsymbol{f}_{\mathrm{ext}}, \tag{S1}$$

in which $\boldsymbol{\sigma}_{\mathrm{ela}}(\boldsymbol{\nabla}\boldsymbol{u})$ and $\boldsymbol{f}_{\mathrm{ext}}(t)$ are the elastic stress and the external driving, respectively, whereas

$$\boldsymbol{f}_{\mathrm{flu}} = \sqrt{2k_{\mathrm{B}}T\bar{\gamma}}\ \frac{\bar{n}}{\sqrt{\bar{g}}}\delta(\xi-1)\partial\boldsymbol{w}$$

is a fluctuating force due to spatially distributed standard white noise $\partial\boldsymbol{w}(\Omega)$ with a solid angle $\Omega = (\theta, \phi)$. For simplicity, we adopt a perfect sphere of unit radius $\bar{R}$ as the equilibrium reference configuration $\bar{\boldsymbol{B}}$, which the body assumes at time $t = 0$. The metric tensor $\bar{\boldsymbol{g}}$ and length of the normal vector $\bar{n}$ then take the form

$$\bar{\boldsymbol{g}}(\xi,\theta,\phi) = \bar{R}^2\,\mathrm{diag}(1, \xi^2, \xi^2\sin^2\theta), \qquad \bar{n} = \bar{R}^2\sin\theta.$$

Note that $\bar{n}/\sqrt{\bar{g}} = \bar{R}^{-1}\xi^{-2}$.

As discussed in the main text (Sec. III), we store the elastic displacement in a co-moving body frame, separately from the global translation $\boldsymbol{A}(t)$ and body-to-laboratory orientation $\mathbf{R}(t)$. At each time step the integrated equation of motion produces a candidate displacement $\boldsymbol{u}(t)+\Delta\boldsymbol{u}$ that we project onto the six global rigid modes of motion parameterized as $\Delta\boldsymbol{A}+\Delta\boldsymbol{\omega}\cdot(\bar{\boldsymbol{B}}-\bar{\boldsymbol{C}})$, where $\Delta\boldsymbol{\omega}$ is antisymmetric and $\bar{\boldsymbol{C}}$ is the centroid of the reference configuration. The residuals define the updated internal displacement $\boldsymbol{u}(t+\Delta t)$. We accumulate the extracted translations and rotation in the laboratory frame as

$$\boldsymbol{A}(t+\Delta t) = \boldsymbol{A}(t) + \mathbf{R}(t)\cdot\Delta\boldsymbol{A}, \qquad \mathbf{R}(t+\Delta t) = \mathbf{R}(t)\cdot\exp(\Delta\boldsymbol{\omega}).$$

In more detail, we consider the body's response to two distinct perturbations of the equilibrium conditions: a point-source force $\boldsymbol{f}_{\mathrm{N}}$ pulling upwards at the body's tip, and a force $\boldsymbol{f}_{\mathrm{S}}$ pushing upwards from below. Specifically, with a bold $\boldsymbol{\Omega}$ denoting the unit radial direction in 3D that corresponds to the solid angle Ω, in the material reference frame the force is applied at the solid angle $\boldsymbol{\Omega}_{\mathrm{ext}}(t) = \mathbf{R}^{\mathrm{T}}(t)\cdot\boldsymbol{\Omega}_{\mathrm{ext}}(0)$ with Cartesian components

$$(f^x_{\mathrm{ext}}, f^y_{\mathrm{ext}}, f^z_{\mathrm{ext}}) = \delta\big(\Omega, \Omega_{\mathrm{ext}}(t)\big)\ \mathbf{R}\cdot(0, 0, z_{\mathrm{ext}}). \tag{S2}$$

where z_{ext} is the force intensity, and $\theta_{\mathrm{ext}}(0) \in \{\theta_{\mathrm{N}}, \theta_{\mathrm{S}}\}$ is the polar angle of the northern $\theta_{\mathrm{N}} = 0$ or southern $\theta_{\mathrm{S}} = \pi$ pole, and $\delta(\Omega, \Omega_{\mathrm{ext}})$ is the surface delta function (Sec. S5).

Equation (S1) integrated over the radial component, can be split into the surface part

$$\bar{\gamma}\int_0^1 d\xi\ \bar{R}\xi^2\frac{\delta(\xi-1)}{\bar{R}\xi^2}\partial_t\boldsymbol{u} = \bar{\gamma}\partial_t\boldsymbol{u}|_{\xi=1} = -\bar{\boldsymbol{n}}\cdot\boldsymbol{\sigma}_{\mathrm{ela}}|_{\xi=1} + \sqrt{2k_{\mathrm{B}}T\bar{\gamma}}\partial\boldsymbol{w} + \frac{z_{\mathrm{ext}}}{2\pi\bar{R}^2\sin\theta}\delta(\theta-\theta_{\mathrm{ext}}), \tag{S3}$$

with the outward normal $\bar{\boldsymbol{n}}$, and the bulk part

$$\boldsymbol{\nabla} \cdot \boldsymbol{\sigma}_{\mathrm{ela}}|_{\xi<1} = 0. \tag{S4}$$

Now we expand Eqs. (S1)–(S4) in a basis set $\psi_{kp}(\xi,\Omega) = P_k(\xi)\chi_p(\Omega)$ formed by combining orthonormal polynomials $P_k(\xi)$ and the *weighted* cardinal functions $\chi_p(\Omega)$. To this end we sample all fields at the joint three-dimensional grid of Gauss-Legendre-Radau radial nodes $\xi_{=1,2,\ldots\Xi}$ and Fibonacci points [1, 2] $\Omega_{p=1,2,\ldots P} = (\theta_p, \phi_p)$ with the weights $w_{np} = w_n\omega_P$ (Secs. S2 and S3). Thereby we can project the displacement field onto the chosen basis functions

$$u^a_{kq} = \langle \psi_{kq}|u^a(\xi,\Omega)\rangle = \sum_{np} w_{np}\psi_{kq}(\xi_n,\Omega_p)u^a(\xi_n,\Omega_p) \approx \int_{\bar{V}} d\bar{V}\ \psi_{kq}u^a,$$

and reconstruct the field values

$$u^a(\xi,\Omega) = \sum_{kq} u^a_{kq}\psi_{kq}(\xi,\Omega).$$

Of course, the surface-bound terms can be represented purely in the basis $\chi_p(\Omega)$. For the external force we have (Sec. S5)

$$(f_{\mathrm{ext}})^a_p = \langle \chi_p|f^a_{\mathrm{ext}}\rangle = f^a_{\mathrm{ext}}\chi_p\big(\Omega_{\mathrm{ext}}(t)\big),$$

with $\Omega_{\mathrm{ext}} = (\theta_{\mathrm{ext}}, 0)$. We approximate the stochastic term of $\boldsymbol{f}_{\mathrm{flu}}(t,\Omega)$ by employing the Karhunen-Loeve expansion with standard temporal white noise components $\dot{W}_p^{a=x,y,z}(t)$ [3]:

$$\partial w^a(\Omega) \approx \sum_p \dot{W}^a_p(t)\chi_p(\Omega), \qquad \left\langle \dot{W}^a_p(t)\right\rangle = 0, \qquad \left\langle \dot{W}^a_p(t)\dot{W}^b_q(s)\right\rangle = \delta(t-s)\delta_{ab}\delta_{pq},$$

which ensures (Sec. S5)

$$\left\langle \partial w^a(t_1,\Omega_1)\partial w^b(t_2,\Omega_2)\right\rangle \approx \delta_{ab}\delta(t_1-t_2)\tilde{\delta}(\Omega_1,\Omega_2).$$

Hence in a time step Δt we get

$$(\delta f_{\mathrm{flu}})^a_p = \int_t^{t+\Delta t} dt'\ (f_{\mathrm{flu}})^a_p(t') = \sqrt{2k_{\mathrm{B}}T\bar{\gamma}}\int_t^{t+\Delta t} dt'\ \left\langle \chi_p\middle|\frac{\delta(\xi-1)}{\bar{R}\xi^2}\partial w^a\right\rangle = \sqrt{2k_{\mathrm{B}}T\bar{\gamma}}\frac{\delta(\xi-1)}{\bar{R}\xi^2}W^a_p(\Delta t),$$

with independent Gaussian random variates $W^a_p(\Delta t)$, such that

$$\left\langle W^a_p(\Delta t)\right\rangle = 0, \qquad \left\langle W^a_p(\Delta t)W^b_q(t+\Delta t)\right\rangle = \Delta t\ \delta_{ab}\delta_{pq}.$$

Both the elastic traction force $\boldsymbol{f}_{\mathrm{ela}} = \bar{\boldsymbol{n}}\cdot\boldsymbol{\sigma}_{\mathrm{ela}}$ and the divergence of the stress in Eq. (S3) and Eq. (S4), respectively, represent linear operators $\mathcal{A}$ that can be in a straightforward way projected onto the basis ψ_{kp} by

$$A_{kq|k'q'} = \langle \psi_{kq}|\mathcal{A}|\psi_{k'q'}\rangle = \sum_{np} w_{np}\psi_{kq}(\xi_n,\Omega_p)\mathcal{A}\psi_{k'q'}(\xi_n,\Omega_p).$$

For instance, with the Lamé coefficient $\bar{\lambda}$ and $\bar{\mu}$ the stress tensor can be cast as

$$\boldsymbol{\sigma}_{\mathrm{ela}} = \begin{pmatrix} \bar{\lambda}\sum_a \partial_a u^a + 2\bar{\mu}\partial_x u^x & \bar{\mu}\left(\partial_y u^x + \partial_x u^y\right) & \bar{\mu}\left(\partial_z u^x + \partial_x u^z\right) \\ \bar{\mu}\left(\partial_y u^x + \partial_x u^y\right) & \bar{\lambda}\sum_a \partial_a u^a + 2\bar{\mu}\partial_y u^y & \bar{\mu}\left(\partial_z u^y + \partial_y u^z\right) \\ \bar{\mu}\left(\partial_z u^x + \partial_x u^z\right) & \bar{\mu}\left(\partial_z u^y + \partial_y u^z\right) & \bar{\lambda}\sum_a \partial_a u^a + 2\bar{\mu}\partial_z u^z \end{pmatrix},$$

In order to expand $\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_{\mathrm{ela}}$, we explicitly write it in terms of the displacements $u^{a=x,y,z}$ using the Kronecker $\boldsymbol{\delta}$:

$$\boldsymbol{\sigma}_{\mathrm{ela}} = \bar{\lambda}\,\mathrm{tr}(\boldsymbol{\epsilon})\boldsymbol{\delta} + 2\bar{\mu}\boldsymbol{\epsilon} = \bar{\lambda}(\boldsymbol{\nabla}\cdot\boldsymbol{u})\boldsymbol{\delta} + \bar{\mu}(\boldsymbol{\nabla}\boldsymbol{u} + \boldsymbol{\nabla}\boldsymbol{u}^{\mathrm{T}}).$$

Then the identity

$$\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_{\mathrm{ela}} = (\bar{\lambda}+\bar{\mu})\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{u}) + \bar{\mu}\boldsymbol{\nabla}^2\boldsymbol{u},$$

in Cartesian coordinates takes the form

$$(\boldsymbol{\nabla}\cdot\boldsymbol{\sigma})^a = (\bar{\lambda}+\bar{\mu})\partial^a \sum_{b=x,y,z} \partial_b u^b + \bar{\mu}\sum_{b=x,y,z}\partial^b\partial_b u^a,$$

where

$$\partial_a = \partial_a\xi\partial_\xi + \partial_a\theta\partial_\theta + \partial_a\phi\partial_\phi = \sum_{i=\xi,\theta,\phi} \mathsf{J}_{ai}\partial_i$$

with the Jacobian J (Sec. S6). Expressing individual differential operators $(\partial_{i=r,\theta,\phi})_{kq|k'q'}$ in the basis ψ_{kq} (Secs. S2 and S3), we can compose the linear operator $(\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}_{\mathrm{ela}})^a_{kq}$ acting on the degrees of freedom u^a_{kq}.

By splitting the degrees of freedom u^a_{kp} into the surface $u_{\mathrm{surf}} = \{u^a_{\Xi p}\}_{a,p}$ and the bulk $u^a_{\mathrm{bulk}} = \{u^a_{k<\Xi,p}\}_{a,k,p}$ parts, we can solve the linear Eq. (S4) for u^a_{bulk} in terms of the surface degrees of freedom (cf. Appendix 11). Then we integrate Eq. (S3) in time by using the Strang-like operator-splitting algorithm [4] with the step size Δt to update the degrees of freedom associated with the surface points u^a_{surf}. A choice of convenient simulation units is described in Sec. S7. While the surface displacement in body coordinates evolves, the bulk displacement is reconstructed from the quasistatic equilibrium condition Eq. (S4). The configuration of the body in the laboratory reference frame is then obtained from Eq. (49) from the main text.

S2 Pseudospectral bases for functions of the radial component

By shifting and orthonormalizing the Jacobi polynomials $P_k^{\alpha,\beta}$ with the shape parameters $\alpha = 0$ and $\beta = 2$, we can express functions of the radial component ξ in the basis

$$P_k(\xi) = (2k+3)^{1/2}P_k^{0,2}(2\xi-1)$$

which satisfies

$$\int_0^1 d\xi\ \xi^2 P_k(\xi)P_{k'}(\xi) = \delta_{kk'}.$$

Furthermore, over the Gauss-Jacobi-Radau quadrature of Ξ points $\xi_{n=1,2,\Xi}$ including the boundary $\xi_\Xi = \bar{R}$, with weights w_n, we have

$$\int_0^1 d\xi\ \xi^2 P_k(\xi)P_{k'}(\xi) = \sum_{n=1}^{\Xi} w_n P_k(\xi_n)P_{k'}(\xi_n) = \delta_{kk'}.$$

Thereby we obtain a pseudospectral basis to expand radial functions as

$$f(\xi) \approx \sum_{k=0}^{\Xi-1} f_k P_k(\xi), \qquad f_k = \sum_{n=1}^{\Xi} w_n P_k(\xi_n) f(\xi_n).$$

The radial derivative can then be defined by a matrix

$$(\partial_r)_{kk'} = \sum_n w_n P_k(\xi_n)\partial_r P'_k(\xi_n).$$

S3 Cardinal functions constructed from Sobolev-Matérn kernels

To approximate functions of the solid angle Ω we construct a Fibanacci grid of P points $\Omega_{p=1,2,...,P} = (\theta_p, \phi_p)$, which covers the sphere nearly uniformly and avoids the poles $\theta \in \{0, \pi\}$. Then we define interpolating cardinal functions $\hat{\chi}_p(\Omega)$ that satisfy the condition $\hat{\chi}_p(\Omega_q) = \delta_{pq}$ and, thus

$$\sum_q \hat{\chi}_p(\Omega_q)\hat{\chi}_{p'}(\Omega_q) = \delta_{pp'}.$$

To this end, we use an expansion of the Sobolev-Matérn spherical kernel [5–7] in the orthonormalized spherical harmonics $Y_{\ell m}(\Omega)$ (Sec. S4) truncated at the order $\ell_{\max}$,

$$K(\Omega_p, \Omega) = \sum_{\ell=0,m=-\ell}^{\ell_{\max},\ell} b_\ell Y_{\ell m}(\Omega_p)Y_{\ell m}(\Omega) = \sum_\ell b_\ell \frac{2\ell+1}{4\pi} P_\ell^0\big(\cos\gamma(\Omega_p, \Omega)\big), \tag{S5}$$

where $b(\ell)$ is the band filter, P_ℓ^0 are the Legendre polynomials, and $\gamma(\Omega_p, \Omega)$ is the *geodesic* distance between the points Ω and Ω_p whose cosine can be evaluated from

$$\cos\gamma(\Omega_p, \Omega) = \cos\theta\cos\theta_p + \sin\theta\sin\theta_p\cos(\phi - \phi_p).$$

The Sobolev-Matérn kernel

$$b_\ell = \left[1 + \frac{4\pi}{P}\ell(\ell+1)\right]^{-(1+s)}. \tag{S6}$$

with the smoothing parameter $s = 3$ is suitable for accurate evaluation of second-order derivatives. As the number of points increases $P \to \infty$ and the band filter tends to the unit ($b_\ell \to 1$), we have

$$K(\Omega, \Omega_0) \to \sum_{\ell=0,m=-\ell}^{\infty,\ell} Y_{\ell m}(\Omega_0)Y_{\ell m}(\Omega) = \frac{1}{\sin\theta}\delta(\theta - \theta_0)\delta(\phi - \phi_0) = \delta(\Omega, \Omega_0).$$

For a sufficiently high order of expansion the matrix $K_{pq} = K(\Omega_p, \Omega_q)$ is invertible, and therefore we can pose

$$\hat{\chi}_p(\Omega) = \sum_q K_{pq}^{-1} K(\Omega_q, \Omega).$$

Furthermore, to frame our numerical methods using scalar products of functions on the sphere we approximate integrals by the sum

$$\int d\Omega f(\Omega) \approx \omega_P \sum_p f(\Omega_p),$$

where the weight $\omega_P = 4\pi/P$ approximates an element of the solid angle associated with each Fibonacci point. Then we define weighted cardinal functions

$$\chi_p = \frac{\hat{\chi}_p}{\sqrt{\omega_P}}.$$

In the basis $\chi_p(\Omega)$ a function $f(\Omega)$ can be interpolated by $f(\Omega) \approx \sum_p f_p\chi_p(\Omega)$ where $f_p = \sqrt{\omega_P}f(\Omega_p) = \omega_P \sum_q \chi_p(\Omega_q)f(\Omega_q)$. In the vector form $\mathsf{f} = (f_1, f_2, ..., f_p)^{\mathrm{T}}$, the partial derivative operator $\partial_{i=\theta,\phi}$ follows from

$$(\partial_i f)_p = \sum_q \chi_p(\Omega_q)\partial_i f(\Omega_q) \approx \sum_q \chi_p(\Omega_q) \sum_{p'} f_{p'}\partial_i\chi_{p'}(\Omega_q) = \sum_{p'} (\partial_i)_{pp'} f_{p'}$$

where $(\partial_i)_{pp'} = \sum_q \chi_p(\Omega_q)\partial_i\chi_{p'}(\Omega_q)$.

S4 Orthonormal real spherical harmonics

To define the cardinal basis (Sec. S3) we employ real spherical harmonics

$$\Upsilon_{\ell m}(\Omega) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{2\pi(\ell+m)!}} P_\ell^m[\cos(\theta)] \begin{cases} \sin(m\phi) \text{ if } m<0, \\ \frac{1}{\sqrt{2}} \text{ if } m=0, \\ \cos(m\phi) \text{ otherwise,} \end{cases}$$

which are orthonormalized so as

$$\int_0^\pi d\theta \ \sin(\theta) \int_0^{2\pi} d\phi \ \Upsilon_{\ell m}(\theta,\phi)\Upsilon_{\ell' m'}(\theta,\phi) = \delta_{\ell\ell'}\delta_{mm'}.$$

Here P_ℓ^m are the associated Legendre polynomials.

S5 Numerical representations of Dirac delta functions

In numerical methods we approximate Dirac delta functions by the truncated series expansions. In the pseudospectral polynomial basis $P_k(\xi)$ (Sec. S2) the radial delta function takes the form

$$\frac{\delta(\xi-\xi_0)}{\bar{R}\xi^2} \approx \sum_k P_k(\xi_0)P_k(\xi). \tag{S7}$$

Note that for any radial function $f(\xi) = \sum_k f_k P_k(\xi)$, we have

$$\int d\xi \ \delta(\xi-\xi_0)f(\xi) \approx \sum_{nkk'} w_n P_k(\xi_0)P_k(\xi_n)f_{k'}P_{k'}(\xi_n) = \sum_{kk'} P_k(\xi_0)\delta_{kk'}f_{k'} = \sum_k f_k P_k(\xi_0) = f(\xi_0). \tag{S8}$$

Aiming for the same property, we define

$$\tilde{\delta}(\Omega,\Omega_0) = \sum_p \chi_p(\Omega_0)\chi_p(\Omega) \approx \delta(\Omega,\Omega_0)$$

which serves as an approximation of the Dirac delta function $\delta(\Omega,\Omega_0)$ centered at the solid angle Ω_0 on the unit sphere. Indeed, for any function of the solid angle $f(\Omega) = \sum_p f_p\chi_p(\Omega)$ we observe that

$$\begin{aligned}\int d\Omega f(\Omega)\delta(\Omega,\Omega_p) \approx \omega_P \sum_p f(\Omega_p)\tilde{\delta}(\Omega_p,\Omega_0) = \omega_P \sum_{pqq'} f_q\chi_q(\Omega_p)\chi_{q'}(\Omega_p)\chi_{q'}(\Omega_0) \\ = \sum_{pqq'} f_q\delta_{pq}\delta_{pq'}\chi_{q'}(\Omega_0) = \sum_p f_p\chi_p(\Omega_0) = f(\Omega_0).\end{aligned} \tag{S9}$$

Note that the weight incorporated in χ_p ensures $\tilde{\delta}(\Omega,\Omega_0) \to \delta(\Omega,\Omega_0)$ as $P\to\infty$.

S6 Jacobian matrix

To evaluate the derivatives of functions parameterized in the spherical coordinates

$$\begin{cases} \xi = \sqrt{x^2+y^2+z^2} \\ \theta = \arctan\left(\frac{\sqrt{x^2+y^2}}{z}\right) \\ \phi = \arctan\left(\frac{y}{x}\right) \end{cases}$$

with respect to the Cartesian components (x, y, z), we define a Jacobian matrix

$$\mathsf{J} = \begin{pmatrix} \partial_x\xi & \partial_y\xi & \partial_z\xi \\ \partial_x\theta & \partial_y\theta & \partial_z\theta \\ \partial_x\phi & \partial_y\phi & \partial_z\phi \end{pmatrix} = \begin{pmatrix} \frac{x}{\xi} & \frac{y}{\xi} & \frac{z}{\xi} \\ \frac{xz}{\xi^2\sqrt{x^2+y^2}} & \frac{yz}{\xi^2\sqrt{x^2+y^2}} & -\frac{\sqrt{x^2+y^2}}{\xi^2} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} & 0 \end{pmatrix} = \begin{pmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \frac{\cos\phi\cos\theta}{\xi} & \frac{\sin\phi\cos\theta}{\xi} & -\frac{\sin\theta}{\xi} \\ -\frac{\sin\phi}{\xi\sin\theta} & \frac{\cos\phi}{\xi\sin\theta} & 0 \end{pmatrix}.$$

S7 Numerical parameters of simulated models

In our numerical examples we adopt data on cell nuclei (Table S1) as a representative system of biological interest [8–16]. Using Eq. (0.1) from Ref. [17, p. xxvii,], we estimate the reference values of the relevant physical parameters form their typical bounds reported in the literature.

Parameter	**Range (SI units)**	**Reference value**	**Source**
Volume of a nucleus $\bar{V}$	122–218 μm^3	200 μm^3	Ref. [16]
Effective radius of a nucleus $\bar{R}$	3–4 µm	3 µm	Eq. (S10)
Mass of a nucleus $\mathcal{M}$	0.1–0.2 pg	0.2 pg	Eq. (S11)
Dynamic viscosity of cytoplasm η	10–100 Pa s	30 Pa s	Refs. [18, 19]
Young modulus E	3–8 kPa	5 kPa	Refs. [9–12]
Poisson ratio ν	0.3–0.5	0.4	Refs. [8–10, 12, 13]
Characteristic time scale τ	2–4 µs	3 µs	Eq. (S14)
Characteristic energy scale ε	90–400 fJ	200 fJ	Eq. (S15)
Dissipative coefficient $\bar{\gamma}$	4–40 kg m^{-2} μs^{-1}	10 kg m^{-2} μs^{-1}	Eq. (S16)
Force generated by a molecular motor z_0	2–8 pN	4 pN	Refs. [20–23]

Table S1: Values of relevant physical parameters for a cell nucleus, cf. Refs. [8–13, 16].

As the unit of length we use $\bar{R}$ derived from the typical volume $\bar{V}$ of a cell nucleus as the radius of an equivalent sphere

$$\bar{R} = \sqrt[3]{\frac{3\bar{V}}{4\pi}}. \tag{S10}$$

We estimate the volume $\bar{V}$ from 3D segmentation of live-imaging data reported in Ref. [16] for nuclei in zebrafish retinal neuroepithelia (Table S1). Because nuclei typically have the density similar to water $\rho \approx 1000\,\mathrm{kg\,m^{-3}}$ [24], their mass can be approximated as

$$\mathcal{M} = \rho\bar{V}. \tag{S11}$$

The Lamé coefficients $\bar{\lambda}$ and $\bar{\mu}$ are derived from the estimates of nuclear Young modulus E and Poisson ratio ν with help of the formulas [25]

$$\bar{\lambda} = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \bar{\mu} = \frac{E}{2(1+\nu)}, \tag{S12}$$

$$E = \frac{\bar{\mu}(3\bar{\lambda}+2\bar{\mu})}{\bar{\lambda}+\mu}, \qquad \nu = \frac{\lambda}{2(\bar{\lambda}+\bar{\mu})}. \tag{S13}$$

The elastic properties of nuclei were measured in Refs. [9–12] through micropipette aspiration, atomic force microscopy, and compression of cells between glass microplates. In particular, the Young modulus is on the order of 5 kPa for nuclei inside the cell and on the order of 8 kPa when isolated.

General bounds of the dimensionless Poisson ratio ν span the interval $[-1, 0.5]$, where the value 0.5 corresponds to the incompressible limit, as adopted in some studies [9, 10, 12]. Allowing for a certain degree of compressibility, we rely on Refs. [8, 13], which measured this parameter by contacting indentation and imaging techniques combined with numerical modeling.

The characteristic time and energy scales, τ and ε respectively, can then be be derived from the elastic modulus as

$$\tau = \sqrt{\frac{\mathcal{M}}{E\bar{R}}}, \tag{S14}$$

$$\varepsilon = E\bar{R}^3. \tag{S15}$$

These values we use the units of time and energy in our simulations.

To evaluate the dissipative coefficient $\bar{\gamma}$, we employ the Stokes law for the friction $\alpha = 6\pi\eta\bar{R}$ where η is the dynamic viscosity of the cytoplasm surrounding nuclei. Refs. [18, 19] report η to be in the range 10–100 Pa s, measured through micropipette aspiration and magnetic spectroscopy. For comparison the dynamic viscosity of water is approximately 1 Pa ms. We thus obtain the dissipative coefficient

$$\bar{\gamma} = \frac{\alpha}{4\pi\bar{R}^2} = \frac{3\eta}{2\bar{R}} \tag{S16}$$

per surface area of the sphere of radius $\bar{R}$.

Our analysis confirms that thermal noise $k_{\mathrm{B}}T_0 = 4\,\mathrm{zJ}$ at $T_0 = 37\,^{\circ}\mathrm{C}$ would produce negligible effects over relevant biological energy scales of nuclei mechanics $\varepsilon \approx 200\,\mathrm{fJ}$. However active processes in cells and their environment typically amplify the level of fluctuations in the system. This behavior is usually modeled by introducing an *effective* temperature T_{eff} [26, 27]. For the sake of illustration, we therefore amplify the fluctuations by a factor of 10^4.

Single molecule experiments show that molecular motors generate forces z_0 in the piconewton range. Kinesin motors exhibit a stall force of 7.2–7.6 pN [20, 21]. For myosin II, optical trapping measurements report forces with a broad distribution spanning 1–7 pN, corresponding to 2.2–4.6 pN to generate forces in the range 3–4 pN [22]. On the other hand, myosin V shows a stall force of 2.7–3.3 pN [23]. In our simulations of the nucleus, we model the localized force of magnitude z_{ext} (Sec. S1) as arising from $N_{\mathrm{m}} \approx 300$ molecular motors, such that

$$z_{\mathrm{ext}} = N_{\mathrm{m}} z_0, \tag{S17}$$

with z_0 in the range of few piconewtons (Table S1). It is worth noticing that in real systems, collective motor behavior can deviate from linear scaling due to complexities such as the ability for individual motors to step in crowded environments, synchronization, load sharing, and stochastic detachment.

The numerical parameter values we used to generate Fig. 2 are summarized in Table S2.

Parameter	Value
First Lamé coefficient $\bar{\lambda}$	$1.4286\,\varepsilon\,\bar{\mathrm{R}}^{-3}$
Second Lamé coefficient $\bar{\mu}$	$0.3571\,\varepsilon\,\bar{\mathrm{R}}^{-3}$
Dissipative coefficient $\bar{\gamma}$	$3\times10^{3}\,\varepsilon\,\tau\,\bar{\mathrm{R}}^{-2}$
Thermal energy $k_{\mathrm{B}}T_0$	$2\times10^{-8}\,\varepsilon$
Effective temperature T_{eff}	$2\times10^{-4}\,\varepsilon/\mathrm{k_B}$
Single motor effective force z_0	$7\times10^{-5}\,\varepsilon\,\bar{\mathrm{R}}^{-1}$
Magnitude of the external force z_{ext}	$2\times10^{-2}\,\varepsilon\,\bar{\mathrm{R}}^{-1}$
Time step Δt	$20\,\tau$
Number of Gauss-Jacobi-Radau points Ξ	10
Number of Fibonacci points P	100
Order of Matérn-Sobolev kernel expansion $\ell_{\max}$	100

Table S2: Parameter values used in our numerical examples.

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